

Design and Functional Simulation of a

Mealy Finite State Machine Based Multi-Product Vending Machine

Using Verilog HDL

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RTL Design — Verilog HDL — Finite State Machines

Table of Contents

1. Introduction	- -	1
(a) Background		
(b) Objectives		
(c) Project Scope		
2. System Overview	- -	1
(a) Features		
(b) Product Specifications		
3. System Design	- -	2
(a) Block Diagram		
(b) Working Principle		
4. Mealy FSM Design	- -	3
(a) Design Selection		
(b) State Definition		
(c) State Transition Diagram		
5. RTL Design	- -	4
(a) RTL Implementation		
(b) Signal Description		
(c) Design Methodology		
6. Functional Verification	- -	6
(a) Test Scenarios		
(b) Simulation Results		
(c) Waveform Analysis		
7. Design Summary	- -	9
(a) Key Features		
(b) Advantages		
(c) Future Enhancements		
8. Conclusion	- -	10

1 Introduction

1.1 Background

A vending machine is an automated system that dispenses products based on user selection and payment. The controller is the core component responsible for processing user inputs, validating transactions, maintaining the balance, dispensing the selected product, and returning change whenever required. Finite State Machines (FSMs) provide a structured and reliable approach for designing such sequential control systems.

This project presents the design and functional simulation of a Mealy Finite State Machine (FSM) based multi-product vending machine using Verilog HDL. The controller supports multiple product selections, accepts different coin denominations, processes confirmation and cancellation requests, and performs automatic change calculation. The complete design is functionally verified through simulation to ensure correct operation under different user interactions.

1.2 Objectives

- Design a multi-product vending machine controller using a Mealy Finite State Machine (FSM).
- Implement the controller using Verilog HDL.
- Support multiple product selections with different product prices.
- Accept multiple coin denominations while maintaining the current balance.
- Provide product confirmation, cancellation, and automatic change return.
- Verify the functional correctness of the design through simulation.

1.3 Project Scope

This project focuses on the RTL design and functional simulation of a Mealy FSM based multi-product vending machine controller using Verilog HDL. The controller supports four products with different prices, two coin denominations, balance tracking, product confirmation, transaction cancellation, and automatic change calculation. The design emphasizes reliable state transitions, accurate transaction handling, and efficient product dispensing while maintaining a simple and modular architecture suitable for digital hardware implementation.

2 System Overview

2.1 Features

The proposed vending machine controller provides the following features:

- Supports four products with different prices.
- Accepts Rs.5 and Rs.10 coin denominations.

- Implements a Mealy Finite State Machine (FSM) for efficient control.
- Maintains the current balance during each transaction.
- Supports product confirmation before dispensing.
- Allows users to cancel the transaction and receive a refund.
- Automatically calculates and returns the appropriate change.
- Functionally verified through simulation using ModelSim.

2.2 Product Specifications

The vending machine supports four products with different selection codes and prices, as listed in Table 1.

Table 1: Product Specifications

Product	Selection Code	Price
Product A	2'b00	Rs. 5
Product B	2'b01	Rs. 10
Product C	2'b10	Rs. 15
Product D	2'b11	Rs. 20

3 System Design

3.1 Block Diagram

Figure 1 illustrates the overall architecture of the proposed multi-product vending machine controller. The controller receives user inputs such as product selection, coin insertion, confirmation, cancellation, and reset signals. Based on these inputs, the Mealy FSM processes the transaction, updates the balance, dispenses the selected product, and returns the appropriate change.

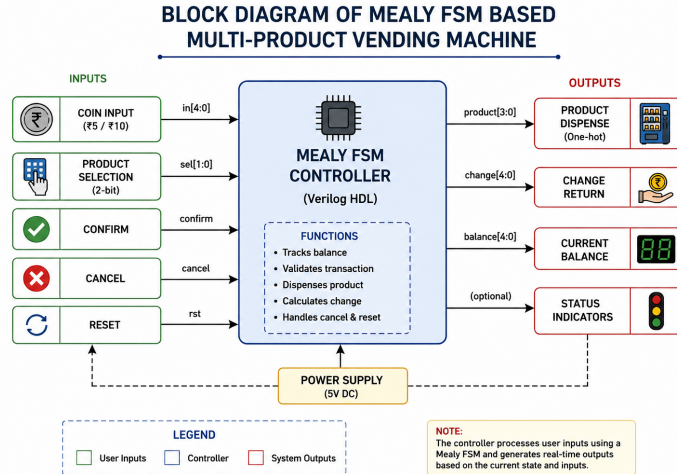


Figure 1: Block Diagram of the Mealy FSM Based Multi-Product Vending Machine

3.2 Working Principle

The proposed vending machine operates by processing user inputs through a Mealy Finite State Machine (FSM). Initially, the controller remains in the idle state with a zero balance. The user begins by selecting one of the available products and inserting valid coin denominations. As coins are inserted, the controller updates the current balance and evaluates whether it is sufficient for the selected product.

Once the required balance is available, the user confirms the transaction. The controller then dispenses the selected product and automatically returns any excess amount as change. If the user cancels the transaction before confirmation, the controller refunds the accumulated balance and returns to the initial state. A reset signal can be used at any time to restore the controller to its default operating condition.

4 Mealy FSM Design

4.1 Design Selection

The vending machine controller is implemented using a Mealy Finite State Machine (FSM), where the outputs depend on both the current state and the current inputs. This approach enables immediate response to user actions such as product confirmation, transaction cancellation, and coin insertion without requiring an additional state transition.

Compared to a Moore FSM, the Mealy model provides faster output generation and a more efficient state representation, making it well suited for vending machine applications that require real-time interaction and reduced hardware complexity.

4.2 State Definition

The vending machine controller consists of four balance states. Each state represents the current balance available in the system and determines the subsequent state transitions based on user inputs and product selection. The state definitions are summarized in Table 2.

Table 2: State Definition

State	Balance	Description
S0	Rs. 0	Initial idle state. Waits for product selection and coin insertion.
S5	Rs. 5	Indicates that the current balance is Rs. 5. Additional coin insertion, confirmation, or cancellation is processed from this state.
S10	Rs. 10	Indicates that the current balance is Rs. 10. The controller evaluates user inputs to determine whether to dispense the selected product or continue the transaction.
S15	Rs. 15	Indicates that the current balance is Rs. 15. The controller can dispense eligible products, return change if required, or process transaction cancellation.

4.3 State Transition Diagram

The state transition diagram shown in Figure 2 illustrates the complete behavior of the proposed Mealy FSM based vending machine controller. The diagram represents all possible state transitions based on coin insertion, product selection, confirmation, cancellation, and reset operations. It also highlights the corresponding output actions such as product dispensing and change return, providing a comprehensive view of the controller's operation.

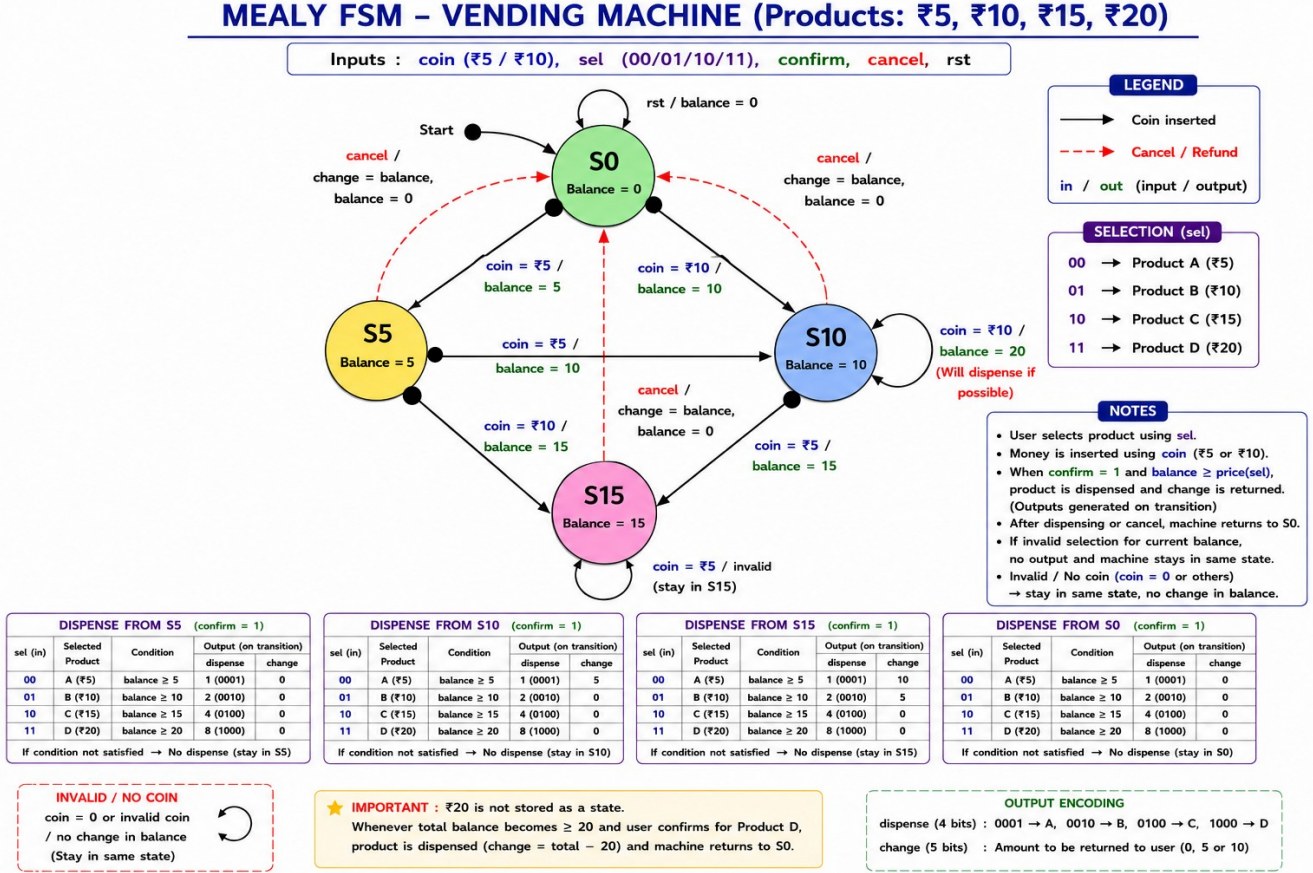


Figure 2: Mealy FSM State Transition Diagram for the Proposed Multi-Product Vending Machine Controller

5 RTL Design

5.1 RTL Implementation

The proposed vending machine controller is implemented using Verilog HDL based on a Mealy Finite State Machine (FSM). The design consists of synchronous sequential logic for state updates and combinational logic for next-state and output generation. The controller processes user inputs, maintains the transaction balance, validates product selection, and controls product dispensing and change calculation.

The implementation follows a modular RTL design approach, ensuring clear separation between state transition logic, balance management, and output generation. This organization improves readability, simplifies functional verification, and allows the design to be synthesized for FPGA or ASIC implementation.

5.2 Signal Description

Table 3 summarizes the input and output signals used in the proposed vending machine controller. These signals facilitate communication between the controller and external components while enabling product selection, transaction processing, and output generation.

Table 3: Signal Description

Signal	Direction	Description
clk	Input	System clock used for synchronous state transitions.
rst	Input	Resets the controller and clears all transaction data.
in[4:0]	Input	Accepts valid coin denominations (Rs. 5 and Rs. 10).
sel[1:0]	Input	Selects one of the four available products.
confirm	Input	Confirms the selected product and enables transaction processing.
cancel	Input	Cancels the current transaction and refunds the available balance.
product[3:0]	Output	Indicates the dispensed product.
change[4:0]	Output	Returns the remaining balance after product dispensing or cancellation.
balance[4:0]	Output	Displays the current balance maintained by the controller.

5.3 Design Methodology

The vending machine controller is developed using a structured RTL design methodology based on a Mealy Finite State Machine (FSM). The design begins by decoding the selected product to determine its corresponding price. Once the product is confirmed, the controller accepts valid coin denominations and updates the current balance accordingly.

As the transaction progresses, the FSM continuously evaluates the accumulated balance against the selected product price. When the available balance is sufficient, the controller dispenses the selected product and automatically calculates any remaining balance as change. If the user chooses to cancel the transaction before product dispensing, the accumulated balance is refunded, and the controller returns to the initial state.

The complete design is implemented using synthesizable Verilog HDL and functionally verified through simulation to ensure correct operation under various transaction scenarios.

6 Functional Verification

6.1 Test Scenarios

The functional behavior of the proposed vending machine controller was verified using a dedicated Verilog testbench in Vivado. The verification process covered both normal operating conditions and boundary cases to ensure reliable controller operation under different user interactions.

The test scenarios were designed to validate product selection, coin insertion, balance accumulation, product dispensing, change calculation, transaction cancellation, and invalid user operations. Table 4 summarizes the major verification scenarios considered during simulation.

Table 4: Summary of Test Scenarios

Test Case	Verification Objective
TC1	Product dispensing with exact payment.
TC2	Product dispensing with excess payment and change return.
TC3	Multiple coin insertion for balance accumulation.
TC4	Verification of different product selections.
TC5	Transaction cancellation and balance refund.
TC6	Coin insertion before product confirmation (invalid operation).
TC7	Insufficient balance for the selected product.
TC8	Invalid coin input verification.
TC9	Reset operation during transaction.

6.2 RTL Schematic

The synthesized RTL schematic of the proposed vending machine controller is shown in Figure 3. The schematic illustrates the hardware representation generated from the Verilog HDL description and confirms the successful synthesis of the design. It also provides a structural view of the controller, including its input and output ports and the internal logic inferred during synthesis.

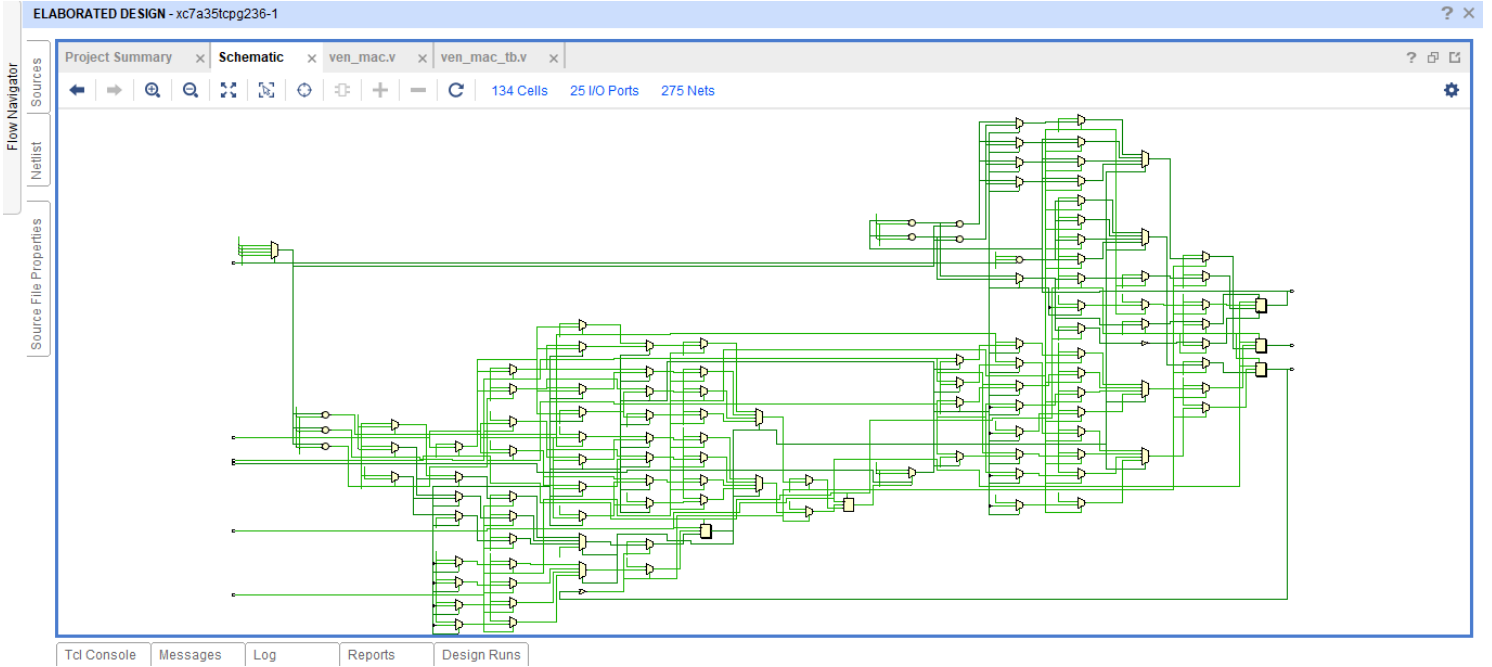


Figure 3: RTL Schematic of the Mealy FSM Based Multi-Product Vending Machine

6.3 Simulation Results

The complete behavioral simulation of the proposed Mealy FSM based vending machine controller is shown in Figure 4. The waveform includes all verification scenarios executed using the developed Verilog testbench in ModelSim. The simulation demonstrates the controller response under different operating conditions, including product selection, coin insertion, confirmation, cancellation, balance update, product dispensing, change return, reset operation, and invalid input conditions.

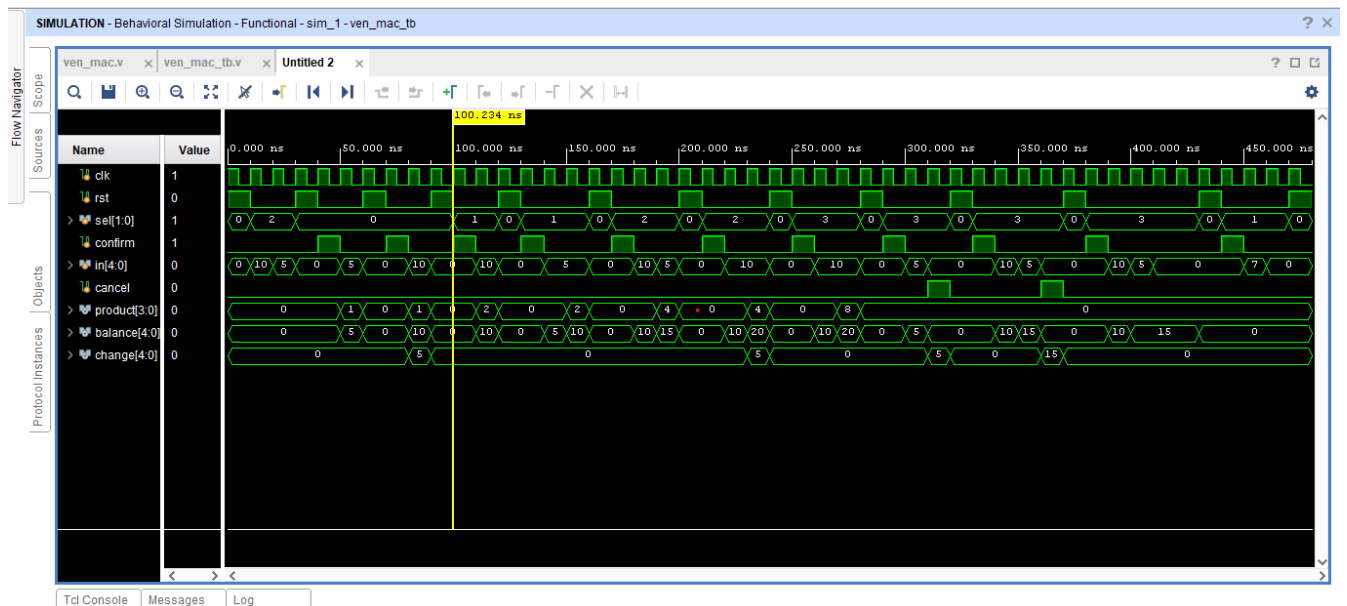


Figure 4: Behavioral Simulation Results of the Proposed Vending Machine Controller

6.4 Waveform Analysis

The simulation results shown in Figure 4 verify the functional correctness of the proposed vending machine controller under different operating conditions. The major verification scenarios are summarized below.

TC0: Coin Insertion Before Confirmation

The user attempts to insert Rs. 10 and Rs. 5 without confirming the selected product. The controller ignores the inputs, and no balance update or product dispensing occurs.

TC1: Exact Payment for Product A

Product A (Rs. 5) is selected and confirmed. After inserting a Rs. 5 coin, the controller dispenses the product without returning any change.

TC2: Excess Payment for Product A

Product A is selected and confirmed. A Rs. 10 coin is inserted, and the controller dispenses the product while returning Rs. 5 as change.

TC3: Exact Payment for Product B

Product B (Rs. 10) is selected and confirmed. The controller dispenses the product after receiving a Rs. 10 coin.

TC4: Multiple Coin Insertion

Product B is selected and confirmed. Two Rs. 5 coins are inserted sequentially, resulting in successful product dispensing.

TC5: Exact Payment for Product C

Product C (Rs. 15) is selected and confirmed. A Rs. 10 coin followed by a Rs. 5 coin results in successful product dispensing.

TC6: Excess Payment for Product C

Product C is selected and confirmed. Two Rs. 10 coins are inserted, and the controller returns Rs. 5 as change after dispensing the product.

TC7: Exact Payment for Product D

Product D (Rs. 20) is selected and confirmed. Two Rs. 10 coins are inserted, and the product is dispensed successfully.

TC8: Transaction Cancellation After Rs. 5

After confirming Product D, the user inserts Rs. 5 and cancels the transaction. The controller refunds the accumulated balance without dispensing any product.

TC9: Transaction Cancellation After Rs. 15

After confirming Product D, the user inserts Rs. 10 and Rs. 5 before cancelling the transaction. The controller returns the full balance and resets the transaction.

TC10: Insufficient Balance

Product D is selected and confirmed. Only Rs. 15 is inserted, which is insufficient to purchase the product. The controller waits for additional payment and does not dispense the product.

TC11: Invalid Coin Input

An invalid coin value is applied to the controller after product confirmation. The controller ignores the invalid input and continues normal operation.

7 Design Summary

7.1 Key Features

The proposed vending machine controller provides the following key features:

- Mealy FSM based controller implemented using Verilog HDL.
- Supports four products with different price levels.
- Accepts Rs. 5 and Rs. 10 coin denominations.
- Maintains the current transaction balance.
- Supports product confirmation before transaction processing.
- Provides transaction cancellation with balance refund.
- Automatically calculates and returns change.
- Functionally verified through ModelSim simulation.

7.2 Advantages

- Simple and modular RTL architecture.
- Efficient state-based transaction control.
- Fast response using a Mealy FSM.
- Easy to understand and maintain.
- Suitable for FPGA and ASIC implementation.
- Easily scalable to support additional products and features.

7.3 Future Enhancements

Future improvements to the proposed vending machine controller include:

- Support for additional coin and currency denominations.
- Integration of digital payment methods such as UPI or NFC.
- LCD or OLED display for user interaction.
- Product inventory monitoring.
- Password-protected maintenance mode.
- FPGA implementation on hardware for real-time validation.

8 Conclusion

A Mealy FSM based multi-product vending machine controller was successfully designed and implemented using Verilog HDL. The controller supports multiple product selections, balance management, product confirmation, change calculation, and transaction cancellation. Functional verification using ModelSim confirmed the correct operation of the proposed design under various test scenarios. The modular RTL architecture and efficient FSM implementation make the controller suitable for digital hardware applications and future enhancements.